Task 6 – Recurrence formulation

For recursive binary search, the original input interval has n keys. Each non-base call makes one recursive call on at most half the interval and performs constant work (middle calculation and comparisons) outside the recursive call. The base case occurs when low > high (empty interval) or a key matches. Therefore $T(n)=T(\lfloor n/2 \rfloor)+c$ for $n>1$, and $T(0)=\Theta(1)$. There is one recursive call per level, not two.